26PYB1002J · UNIT 3 · QUANTUM PHYSICS

When particles behave like waves

Matter waves, the uncertainty principle, the wave of probability and Schrödinger's equation: Lectures 3, 6, 7 and 8

THE ROAD MAP

Three lectures, one story

LECTURE 3

Matter has a wave

$\lambda = h / p$

→

LECTURE 6

Waves are spread out

$\Delta x \cdot \Delta p \geq \hbar/2$

→

LECTURE 7

The wave is probability

$|\psi|^2$ and proof

An electron is **neither a tiny ball nor a water wave**. It is a quantum object that shows wave behaviour in some experiments and particle behaviour in others.

03

Matter waves

If light can be a particle, can a particle be a wave?

LECTURE 3 · MATTER WAVES

Light already plays both roles

As a wave

Interference and diffraction: light spreads, overlaps and cancels.

As a particle

Photoelectric and Compton effects: light arrives in packets (photons).

1924 · LOUIS DE BROGLIE

Nature loves symmetry, so matter should be dual too. The waves of moving particles are called **matter waves**.

LECTURE 3 · MATTER WAVES

From light to matter in three steps

STEP 1

One photon, two descriptions

$E = h\nu$

$E = pc$

STEP 2

Equate, and use $c = \nu\lambda$

$p = h\nu / c$

$p = h / \lambda$

STEP 3 · THE LEAP

Apply it to every particle

$\lambda = h / p$

$\lambda = h / mv$

Watch the symbols: ν (nu) is frequency, v is velocity.

LECTURE 3 · MATTER WAVES

Why doesn't a cricket ball diffract?

Object	Mass, speed	$\lambda = h / mv$	Compared with
Cricket ball	0.16 kg, 42.83 m/s	$\approx 1 \times 10^{-34}$ m	10^{20} times smaller than a nucleus: invisible
Electron	9.11×10^{-31} kg, 10^{6} m/s	7.3×10^{-10} m	about the spacing of atoms: observable

Wave effects show up only when λ is comparable to the size of what the wave meets.

LECTURE 3 · MATTER WAVES

One formula, four disguises

The trick every time: **$p^2 = 2m \times$ (kinetic energy)**, then $\lambda = h / p$.

From momentum

$$\lambda = h / mv$$

From kinetic energy E

$$\lambda = h / \sqrt{(2mE)}$$

From voltage V (KE = eV)

$$\lambda = h / \sqrt{(2meV)}$$

From temperature T (KE = 3/2 kT)

$$\lambda = h / \sqrt{(3mkT)}$$

LECTURE 3 · FOR ELECTRONS ONLY

The number to remember

Put h, m and e of the electron into h / √(2meV):

$$\lambda = 12.27 / \sqrt{V}\ \text{Å}$$

V in volts · 1 Å = 10^{-10} m

Example: V = 54 volts

$\lambda = 12.27 / 7.35$

$$\approx 1.67\ \text{Å}$$

Keep this number. In Lecture 7, an experiment at exactly 54 V measures 1.65 Å.

LECTURE 3 · CHECK YOUR UNDERSTANDING

Three quick questions

1. A proton and an electron move at the same speed. Which has the longer de Broglie wavelength?
2. You double an electron's accelerating voltage. What happens to λ?
3. Neutrons have no charge. Why can they still be diffracted by a crystal?

06

The uncertainty principle

If a particle is a wave, where exactly is it?

LECTURE 6 · UNCERTAINTY

Sharp wavelength or sharp position: never both



A long, regular wave

λ is clear, so p = h/λ is known. But where is the particle? Anywhere along it.



A short wave packet

Position is clear. But a short burst mixes many wavelengths, so p is spread out.

This is not a flaw in our instruments. It is built into being a wave.

LECTURE 6 · UNCERTAINTY

Heisenberg's uncertainty principle

$$\Delta x \cdot \Delta p \geq \hbar/2 \text{ where } \hbar = h/2\pi$$

Position and momentum

Squeeze Δx and Δp must grow. Halve Δx and the minimum Δp doubles.

Energy and time: $\Delta E \cdot \Delta t \geq \hbar/2$

A state that lasts a short time has a fuzzy energy, so spectral lines have a natural width.

Books write ≥ h, ≥ ħ or ≥ ħ/2. The exact limit is ħ/2; for quick estimates we use Δx · Δp ≈ ħ.

LECTURE 6 · UNCERTAINTY

Why you never notice it

A 1 g bead, located to 1 μm

$$\Delta v \approx 10^{-25}\ \text{m/s}$$

Completely unnoticeable

An electron, located to 1 Å

$$\Delta v \approx 10^{6}\ \text{m/s}$$

Over 1,000 km every second

h is tiny, so uncertainty only matters for tiny things.

LECTURE 6 · APPLICATION

Can an electron live inside the nucleus?

Step	Result
1. Confine it to the nucleus (diameter)	$\Delta x \approx 2 \times 10^{-14}$ m
2. Uncertainty estimate: $\Delta p \approx h / (2\pi\Delta x)$	$\Delta p \approx 5.27 \times 10^{-21}$ kg m/s, so $p \approx \Delta p$
3. Kinetic energy: $K = p^2 / 2m$	$K \approx 1.52 \times 10^{-11}$ J ≈ 96 MeV
4. Compare with β-decay electrons	only about 4 MeV observed

Energy needed (96 MeV) is about 24× more than observed (4 MeV), so **electrons cannot exist inside the nucleus.**

SO WHERE DO B-DECAY ELECTRONS COME FROM?

They are created at the moment of decay.

$$n \rightarrow p + e^{-} + \bar{\nu}$$

A neutron turns into a proton, an electron and an antineutrino. The electron never lived inside the nucleus.

LECTURE 6 · CHECK YOUR UNDERSTANDING

Three quick questions

1. You halve Δx. What happens to the minimum Δp?
2. Would a perfect instrument beat the uncertainty principle? Why or why not?
3. An excited state lasts 10^{-8} s. Can its energy be perfectly sharp?

07

The wave function and its proof

What is actually waving, and how do we know electrons are waves?

LECTURE 7 · BORN INTERPRETATION (1926)

What is waving? Probability.

$$\text{probability density} = |\psi|^2 = \psi^*\psi$$

- ψ itself can't be measured. It may be negative or even complex.
- Squaring it gives how likely the particle is to be found at each place.
- Borrowed from light: intensity ∝ (amplitude)².

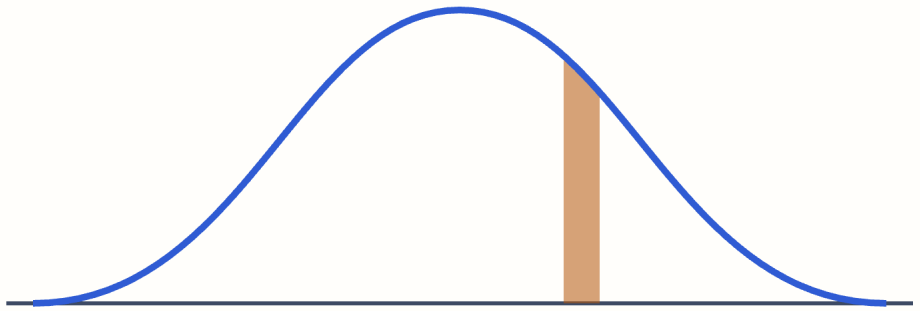


Like a rain-chance map

It gives the odds at each place, not where each drop will land. |ψ|² is the electron's chance map.

LECTURE 7 · BORN INTERPRETATION

Probability is an area under $|\psi|^2$



Blue curve: $|\psi(x)|^2$. Orange strip: width dx.

$$P(x \text{ to } x + dx) = |\psi(x)|^2 \, dx$$

- High curve: likely. Zero: never.
- Between a and b: the area, $\int |\psi|^2 \, dx$
- At exactly one point? Zero. Always ask about an interval.

LECTURE 7 · PHYSICAL SIGNIFICANCE OF Ψ

What makes ψ acceptable

Single-valued

One probability at each point in space and time.

Continuous

No sudden jumps from one place to the next.

Finite

Never infinite anywhere.

Normalised

Total probability over all space is 1: $\iiint |\psi|^2 \, d\tau = 1$

ψ is usually **complex**, so it can't be measured directly. Only $|\psi|^2$ can.

LECTURE 7 · NORMALISATION

Choose A so the total is 1

Example: $\psi = A \sin(\pi x/L)$ for $0 < x < L$, zero elsewhere

$$\int_0^L A^2 \sin^2(\pi x/L)\, dx = 1$$

$$A^2 \cdot L/2 = 1$$

$$\mathbf{A = \sqrt{(2/L)}}$$

Why $\psi^*\psi$, not ψ^2?

ψ may be complex. If $\psi = a + ib$, then

$$\psi^*\psi = a^2 + b^2$$

always real and never negative.

LECTURE 7 · SECTION

Verification of matter waves

de Broglie's idea was a hypothesis. Here's the experiment that proved it.

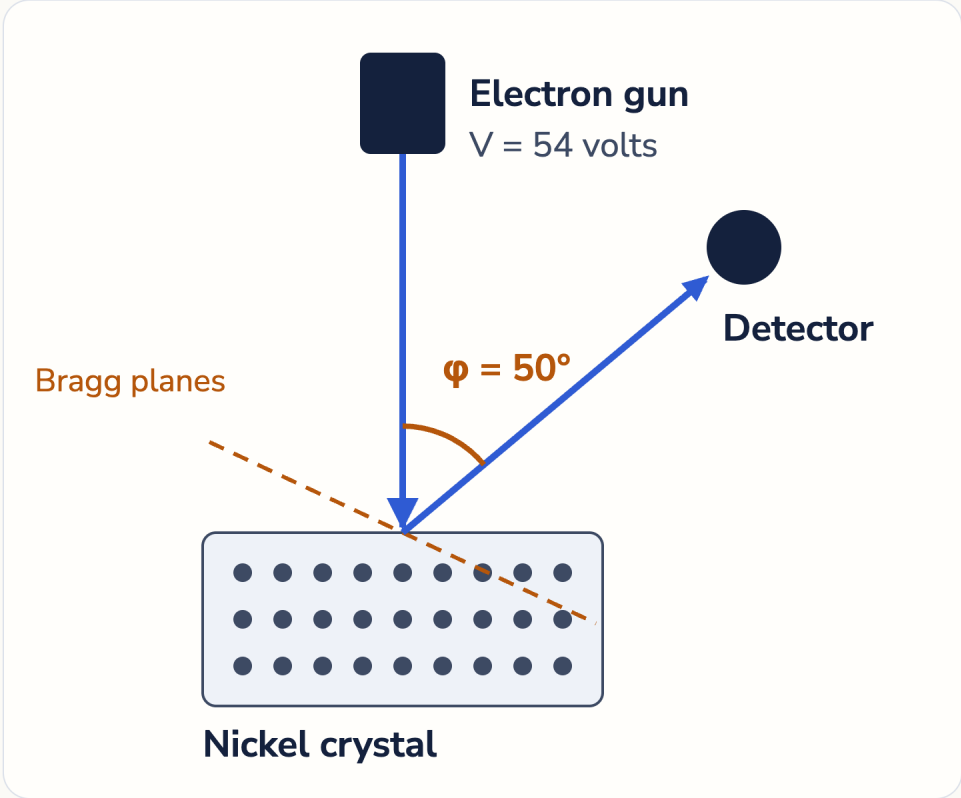
Recall from Lecture 3: at 54 V

$\lambda = 12.27 / \sqrt{54}$

≈ 1.67 Å

LECTURE 7 · DAVISSON–GERMER (1927)

Firing electrons at a crystal



1. **Gun:** a heated filament gives electrons; voltage V speeds them up and sets λ.
2. **Nickel crystal:** rows of atoms about 1 Å apart act as a grating.
3. **Detector:** rotates to measure how many electrons come off at each angle.
4. **Result:** a sharp peak at φ = 50° when V = 54 V.

LECTURE 7 · THE PUNCHLINE

Measured and predicted agree

Measured, with Bragg's law

$2d \sin\theta = n\lambda$

$d = 0.91$ Å, $\theta = 65°$, $n = 1$

1.65 Å

Predicted, by de Broglie

$\lambda = 12.27 / \sqrt{V}$ Å

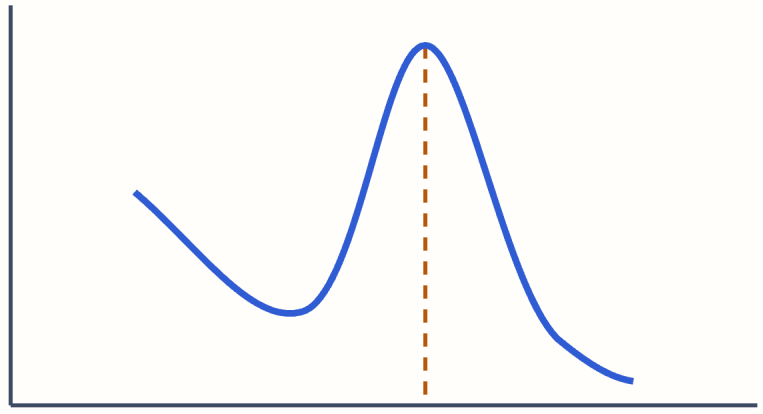
$V = 54$ V

1.67 Å

$\theta = 65°$ because $\theta + 50° + \theta = 180°$: the Bragg planes split the angle between the beams.

LECTURE 7 · DAVISSON–GERMER

Why a peak proves waves



Intensity vs angle at 54 V: peak at 50°

- **Bouncing balls** would give roughly the same count at every angle.
- **Observed:** a sharp peak at 50°, the fingerprint of constructive interference.
- Each electron lands as one dot (particle); the dots pile up where $|\psi|^2$ is large (wave).

J. J. Thomson (Nobel 1906): the electron is a particle. His son G. P. Thomson (Nobel 1937): it is a wave. Both were right.

LECTURE 7 · CHECK YOUR UNDERSTANDING

Three quick questions

1. $\psi = -0.3$ at some point. Is that allowed? What is $|\psi|^2$ there?
2. Why must every wave function be normalised?
3. In Davisson–Germer, what happens to the 50° peak if V changes?

08

Schrödinger's wave equation

How do we actually find ψ? The “F = ma” of quantum mechanics.

LECTURE 8 · SCHRÖDINGER'S EQUATION

Three ingredients, one equation

A CLASSICAL WAVE

$$\nabla^2\psi = (1/v^2)\,\partial^2\psi/\partial t^2$$

+

DE BROGLIE (L3)

$$\lambda = h / mv$$

+

ENERGY

$$E = \tfrac{1}{2}mv^2 + V$$

RESULT · SCHRÖDINGER, 1926

An equation that tells the matter wave ψ what shape to take, just as $F = ma$ tells a ball how to move.

LECTURE 8 · SCHRÖDINGER'S EQUATION

The derivation in four steps

1 · REMOVE TIME

$$\psi = \psi_0 e^{(-i\omega t)} \Rightarrow \partial^2\psi/\partial t^2 = -\omega^2\psi$$

2 · USE Ω/V = 2Π/Λ

$$\nabla^2\psi + (4\pi^2/\lambda^2)\psi = 0$$

3 · PUT IN Λ = H/MV

$$\nabla^2\psi + (4\pi^2 m^2 v^2/h^2)\psi = 0$$

4 · USE M²V² = 2M(E − V)

$$\nabla^2\psi + (8\pi^2 m/h^2)(E - V)\psi = 0$$

Careful: v in step 2 is the wave's speed; v in step 3 is the particle's speed.

LECTURE 8 · THE RESULT

Time-independent Schrödinger equation

In three dimensions

$$\nabla^2\psi + (2m/\hbar^2)(E - V)\psi = 0$$

In one dimension

$$d^2\psi/dx^2 + (2m/\hbar^2)(E - V)\psi = 0$$

$\hbar = h/2\pi$ · E = total energy · V = potential energy · E − V = kinetic energy

LECTURE 8 · READING THE EQUATION

It's energy conservation, written for a wave

Rearranged

$$-(\hbar^2/2m)\, d^2\psi/dx^2 + V\psi = E\psi$$

kinetic + **potential** = **total** energy

“Time-independent”: V depends only on position, so $|\psi|^2$ doesn't change in time.

Check: a free particle (V = 0)

$$\psi = A \sin(kx)$$

$$k = \sqrt{(2mE)} / \hbar = p / \hbar$$

So $\lambda = 2\pi/k = h/p$: **de Broglie comes straight back out.**

LECTURE 8 · CHECK YOUR UNDERSTANDING

Three quick questions

1. In the equation, what do E, V and E – V stand for?
2. For a free particle (V = 0), ψ = A sin(kx). What is k?
3. Why is it called the “time-independent” equation?

WRAP-UP

Unit 3 on one slide

Lecture	Big idea	Key formula
3	Every moving particle has a wave	$\lambda = h / p = h / \sqrt{(2mE)}$
6	A spread-out wave can't have sharp x and p	$\Delta x \cdot \Delta p \geq \hbar/2$
7	$\lvert\psi\rvert^2$ is probability, and electrons diffract	$\iiint \lvert\psi\rvert^2 \, d\tau = 1$; 1.65 Å ≈ 1.67 Å
8	Schrödinger's equation tells ψ what shape to take	$d^2\psi/dx^2 + (2m/\hbar^2)(E - V)\psi = 0$

Particles travel like waves and arrive like particles.